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Lessons

Solving Word Problems with Systems

Systems are very helpful in solving word problems because you can use more than one variable to represent the information in the problem.

EXAMPLE

The sum of two numbers is 15. Nine times the first number is 3 more than twice the second. What are the two numbers?

Step 1

Write a system to represent the information in the problem. You can use x and y for the two numbers.

Written using words	Written using algebra
The sum of two numbers is 15.	$x + y = 15$
Nine times the first number is 3 more than twice the second.	$9x = 3 + 2y$

Write the system: $\begin{cases} x + y = 15 \\ 9x = 3 + 2y \end{cases}$

Step 2

Choose a method for solving the system. It will be easy to isolate a variable in the first equation, so substitution is the best method.

Step 3

Solve.

Isolate a variable.

$x + y = 15$
$y = 15 - x$

Substitute $15 - x$ for y in the other equation.

$9x = 3 + 2y$
$9x = 3 + 2(15 - x)$
$9x = 3 + 30 - 2x$
$9x = 33 - 2x$
$11x = 33$
$x = 3$

Substitute the value for x into the first equation.

$x + y = 15$
$3 + y = 15$
$y = 12$

The numbers are **3 and 12**.

Look back. Read the words of the problem again and see whether your answers make sense.

The sum of two numbers is 15. 3 + 12 = 15 True

Nine times the first number is 3 more than twice the second.

9(3) = 3 + 2(12)
27 = 3 + 24
27 = 27 True

Writing a system is another tool that you can use to solve problems. Good test takers know that there are many ways to solve a problem. Don’t worry about finding the perfect way. Use a method that makes sense to you.

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Lesson 4: Systems of Equations

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